Week 17 Update: Quality Metrics Evaluation (ISO/IEC 25010)

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The chatbot scores 96.8 of 100 (Excellent) across the eight ISO/IEC 25010 quality characteristics on the current build, measured by the system's own Quality scorecard. This update adds the eight characteristics to Section 3 and replaces the Oct 28 numbers in 3.1.

3.8 The eight ISO/IEC 25010 characteristics

Each characteristic is scored 0 to 100 from its checks in the system's Quality scorecard (Admin panel, Quality tab). Scores come from a run on this build against a local test database. The live site shows its own values in the same tab.

#	Characteristic	Score	What is checked	Evidence on this build	Limit
1	Functional Suitability	86.6	Completeness, correctness, appropriateness	13 of 13 topics have 15 or more training questions; held-out accuracy 73.2% of 71 questions (macro F1 0.77); 14 of 14 chat answers placed in a topic; 6 of 6 core routes registered	Correctness is held back by a small synthetic dataset; user helpfulness (thumbs) needs 5 ratings and has 0
2	Performance Efficiency	100	Time behaviour, resource use	Model answer p95 0.63 ms over 120 questions; database round trip p95 0.33 ms over 60 queries; model file 0.9 MB; 338 KB of own scripts and styles	Local run with no load test; live chat timing needs 5 requests

#	Characteristic	Score	What is checked	Evidence on this build	Limit
3	Compatibility	100	Interoperability, coexistence	4 of 4 export routes (CSV, JSON, Orange .tab); 12 of 12 pages have a viewport and HTML5 doctype; no third-party downloads; any PostgreSQL via DATABASE_URL; each role can have its own address	Phone layout is a check on markup, not a test on a real phone
4	Interaction Capability	100	Learnability, accessibility, error protection	12 of 12 pages declare a language; 13 of 13 images have alt text; 58 ARIA attributes; reduced motion, dark mode and text size; guided tutorial and Ctrl+K palette; 46% of training questions are Taglish or Filipino	Usability by students (System Usability Scale) has 0 answers and needs 5; no screen-reader audit
5	Reliability	100	Availability, fault tolerance, recoverability	Database answered in 1 to 2 ms; a full backup was downloaded the same day (0 days ago); model refuses to load if its dataset hash differs; offline banner, idle sign-out, closable portal	The backup check passes only while the last backup is under 30 days old, so it must be repeated monthly; the server error rate needs 20 requests to be judged
6	Security	91.7	Confidentiality, integrity, accountability	6 of 6 security headers; Content-Security-Policy refuses inline scripts; PINs hashed with salted PBKDF2 (6 of 6); lock-out after 5 wrong PINs; audit log of sign-ins; optional admin two-factor code (TOTP)	Sign-in protection scored 0.5 here because the two-factor code and a fixed SECRET_KEY were not set; set them on the live site

#	Characteristic	Score	What is checked	Evidence on this build	Limit
7	Maintainability	95.7	Modularity, testability, reusability	503 automated tests in 26 files, all passing; GitHub Actions runs them on every pull request; 7 of 7 dependencies pinned; model rebuilt by one command	Chatbot.py is 2,273 lines (modularity 0.69); 48% of functions have a docstring
8	Flexibility	100	Adaptability, installability, scalability	18 settings read from environment variables; runs on Vercel, Docker or plain Flask; retrain, feedback-to-dataset and migration scripts; English, Filipino and Taglish in one model; no server state	Scale-out is declared by design, not load tested

Overall: 96.8 of 100, grade Excellent, using the ISO/IEC 25010:2023 names. That edition lists a ninth characteristic, Safety (harm to people or property), which is left out because this web system has no physical hazard. Downloading a full backup raised Reliability from 80.0 to 100 and the overall from 94.2 to 96.8; no other score changed.

3.1 Summary, refreshed

The Oct 28 figures (404 tests, 180 rows, 55.6% accuracy) are out of date. These replace them.

Measure	Oct 28 build	Now	Source
Automated tests	404 passed	503 passed (about 90 s)	unit test suite
Dataset	180 rows, 12 intents of 15	351 rows, 13 topics of 27 (adds greeting and 156 new questions)	property_management_dataset.csv
Split	144 train, 36 test	280 train, 71 test	metrics.json

Measure	Oct 28 build	Now	Source
Held-out accuracy	55.6%	77.5% raw; 73.2% counting the fallback	train_model.py
Held-out macro F1	0.540	0.773 raw; 0.776 counting the fallback	train_model.py
Turned away by the fallback	3 of 36	9 of 71	train_model.py
Quality scorecard	not available	96.8 of 100	Admin, Quality tab

The two splits are different questions, so the rise is a direction, not an exact gain. The 0.70 macro F1 target proposed in 1.5 is now met on the held-out set, but the dataset is still synthetic.

3.9 How the scores are made, and their limits

Every check carries a label so a reader can tell proof from claim.

- **Measured**: a number taken while the scorecard runs (timings, route counts, hashed PINs, security headers).
- **Inspected**: read from the code or files (alt text, pinned packages, tests found, held-out results from the last training run).
- **Declared**: a design fact the scorecard cannot test (lock-out times, scale-out by design).
- A check with too little data shows "needs 5 to judge" and is left out of the score instead of counting as a pass.

What these numbers do not prove:

- The 96.8 is the team's own rubric against the ISO/IEC 25010 characteristics. It is not a certification, and the weights are our choice.
- Accuracy uses a small synthetic dataset written by the team. Real student questions, collected through the Feedback export, are the next evidence.

- Usability has no student score yet. The System Usability Scale survey (Settings, Rate this assistant) needs 5 answers, which the User Acceptance tests in Section 1.11 can collect.
- Performance was timed on one local machine with no load. Section 3.6 (not measured) still applies to load, accessibility audits and real-phone tests.
- Pull request 31 (two-factor sign-in, strict CSP, the 351-row dataset) is live on main. Pull request 34 (one-word topic rules and the Overview portal switch) is not merged yet; these scores do not depend on it. Roles in Sections 1 and 2 that mention a guard no longer match the current app, which has admin, coordinator and student.

Where to see it live: Admin panel, Quality tab, and the printable page at /admin/quality-report.